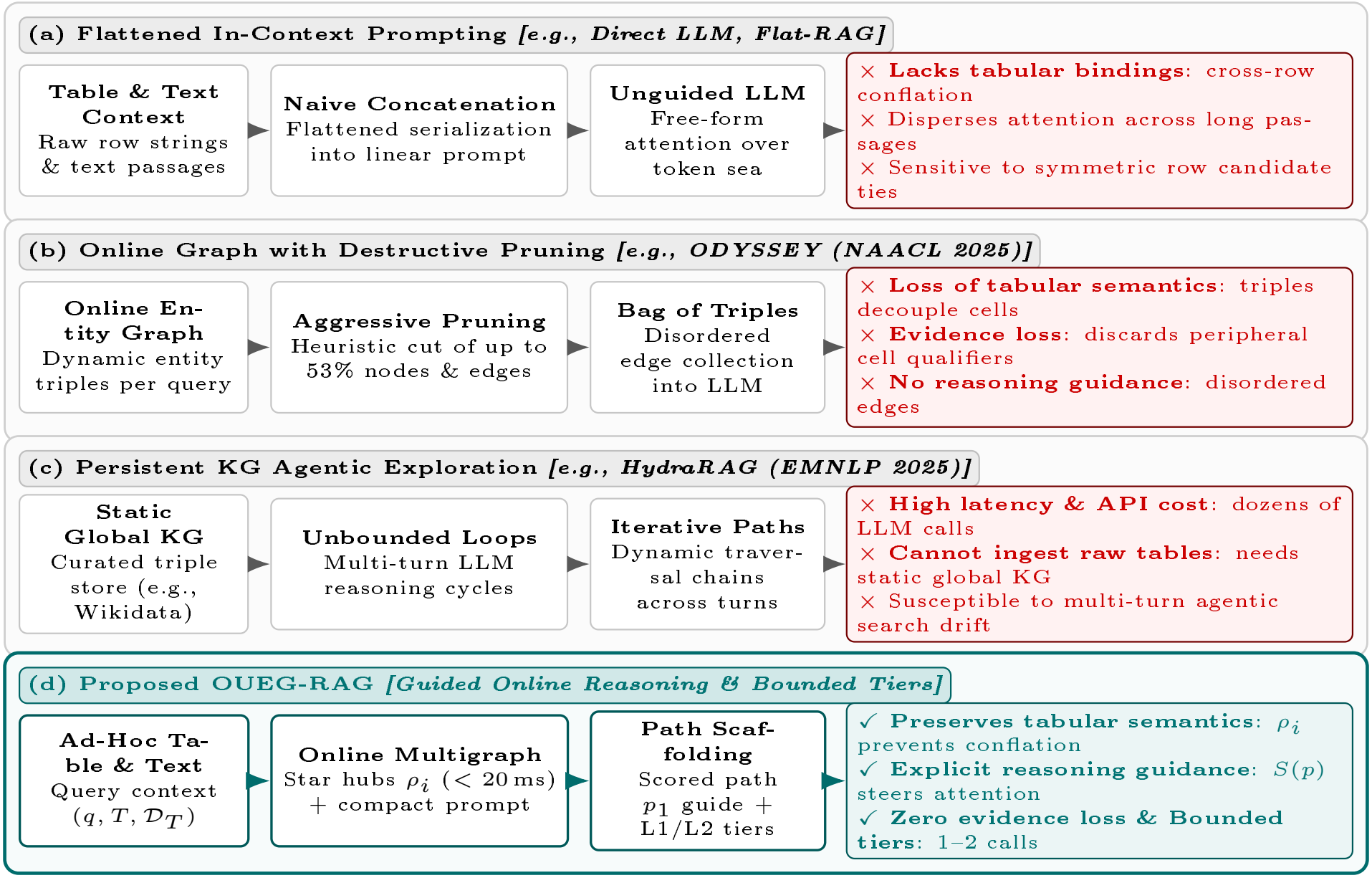

(a) Flattened In-Context Prompting [e.g., Direct LLM, Flat-RAG]
Table & Text Context
Raw row strings & text passages
Naive Concatenation
Flattened serialization into linear prompt
Unguided LLM
Free-form attention over token sea
× Lacks tabular bindings: cross-row conflation
× Disperses attention across long passages
× Sensitive to symmetric row candidate ties
(b) Online Graph with Destructive Pruning [e.g., ODYSSEY (NAACL 2025)]
Online Entity Graph
Dynamic entity triples per query
Aggressive Pruning
Heuristic cut of up to 53% nodes & edges
Bag of Triples
Disordered edge collection into LLM
× Loss of tabular semantics: triples decouple cells
× Evidence loss: discards peripheral cell qualifiers
× No reasoning guidance: disordered edges
(c) Persistent KG Agentic Exploration [e.g., HydraRAG (EMNLP 2025)]
Static Global KG
Curated triple store (e.g., Wikidata)
Unbounded Loops
Multi-turn LLM reasoning cycles
Iterative Paths
Dynamic traversal chains across turns
× High latency & API cost: dozens of LLM calls
× Cannot ingest raw tables: needs static global KG
× Susceptible to multi-turn agentic search drift
(d) Proposed OUEG-RAG [Guided Online Reasoning & Bounded Tiers]
Ad-Hoc Table & Text
Query context $(q, T, \mathcal{D}_T)$
Online Multigraph
Star hubs ρ_i (< 20 ms) + compact prompt
Path Scaffolding
Scored path p_1 guide + L1/L2 tiers
✓ Preserves tabular semantics: ρ_i prevents conflation
✓ Explicit reasoning guidance: $S(p)$ steers attention
✓ Zero evidence loss & Bounded tiers: 1–2 calls